# CHAPTER 1

# INTRODUCTION

## 1.1 Overview

Machine learning is a branch of computer science that researches algorithms and strategies for eliminating manual and complicated issues that are difficult to solve using traditional approaches. (Rebala, Ravi, & Churiwala, 2019) Machine learning focuses on analyzing data and advances to replicate how individuals enhance and develop accuracy over time. (IBM Cloud Education, 2020) Natural language processing (NLP) is a set of theory-driven computing approaches for automatically analyzing and representing human languages. (Chowdhary, 2020) It is a field of machine learning focused on teaching computers to interpret text and spoken language in the same way that people do. NLP integrates computational linguistics modeling of public language with analytical, machine learning, and deep learning models. (IBM Cloud Education, 2020)

A chatbot is a program of a form of communication that enables computers to talk and converse in the same way that humans do. Initially, a chatbot can only respond to conventional inquiries whose questions and responses have been pre-programmed into the system. It's a new type of customer service that uses artificial intelligence and a chat interface. (Nuruzzaman & Hussain, 2018) With technical advancements, machines can eventually answer a freelanced question like a human by passing the Turing Test, which is a test of intellect that is closer to human intelligence. (Oppy, Dowe, & David, 2021) With the rapid advancement of artificial intelligence in recent years, intelligent chatbots have entered a new era and are now widely used in various businesses. The computer performs Natural Language Processing (NLP) to overcome this obstacle. It is the process of identifying the input as well as the correct meaning of a specific action or need because of the occurrence of unknown and unexpected components in the information and the need to determine the proper understanding to apply to do it. Machine learning is used by intelligent chatbots to know and understand from users' requirements and messages. Intelligent chatbots are programmed to recognize terms and concepts with unique meanings that prompt a response. They gradually taught themselves to grasp additional queries to provide helpful responses.

Many Companies and Hospitals are utilizing NLP and machine learning to create bots that can have natural conversations indistinguishable from human talks. Companies focus on creating functional bots that can swiftly and efficiently complete specified tasks. The problem is that maintaining perspective, distinguishing fact from fiction, and managing user expectations can be challenging with artificial intelligence. While AI, machine learning, and natural language processing are frequently mentioned when discussing chatbots, many highly efficient bots are not productive and do not appear human. In this study, we are going to train an intelligent chatbot more accurately and efficiently by using natural language processing. Thus it can make conversation with the people to help them precisely as they want help from a service center or a medical support center.

## 1.2 Background

Today's industry and study require the use of machine learning. Algorithms and neural network models are used to aid machines in increasing productivity over time. Without being specifically trained to make those decisions, Machine Learning algorithms simultaneously develop a numerical computation utilizing the instance of info – often referred to as "training data" – to create choices. To some extent, machine learning is designed on the basis of brain cell connections. In his book The Organization of Behavior, Donald Hebb developed the paradigm, published in 1949. Hebb's perspectives on neuron stimulation and neural interaction are presented in this book. When applied to artificial neural networks and artificial neurons, Hebb's model can be described as a method of changing the interactions between artificial neurons (also known as nodes) and the changes in individual neurons. When two neurons/nodes are triggered at the exact moment, their link strengthens; when they are activated separately, it weakens. The term "weight" is used to describe these interactions, and nodes/neurons with substantial positive weights have both positive and negative tendencies. Negative consequences develop in nodes that tend to have opposite weights (KLOPF, 2013).

By the classification of traditional machine learning, an algorithm improves its forecast accuracy over time. Supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning are the four primary methodologies.

**Supervised learning:** Data scientists supply identified training sets to systems and indicate the elements they need the software to seek for correlations here between this type of machine learning. The algorithm's measured values are both provided. (Taiwo Oladipupo Ayodele, 2010)

**Unsupervised learning:** Machines that learn on unlabeled data are used in this sort of machine learning. The application searches for significant connections between data sets. All of the info required to train algorithms is preset, as are the estimates or suggestions they produce. (Taiwo Oladipupo Ayodele, 2010)

**Semi-supervised learning:** The two preceding methodologies are combined in this machine learning strategy. Despite the fact that data scientists could provide a model with widely labeled data, the model is able to pursue the data and develop a knowledge of the set. (Taiwo Oladipupo Ayodele, 2010).

**Reinforcement learning:** Reinforcement learning is a technique used by data professionals to teach a machine to complete a number of operations with well-defined criteria. Data scientists develop an algorithm to execute a job and offer valuable feedback as the system learns how to do so. However, the algorithm chooses which steps to perform together with the way by itself. (Taiwo Oladipupo Ayodele, 2010)

A chatbot, as previously said, is software with a specific amount of artificial intelligence that talks with a person or another chatbot to give the viewer of the interaction the projection that they are conversing with a natural person. Users may unwittingly integrate this technology into societal conventions, associations, and responsibilities as if they were actual persons. Programmers primarily focus on two factors while creating chatbots: emotions and agency. With the tremendous spread of the Internet, particularly social networking sites, the use of chatbots is on the rise, whether with a simple or more powerful artificial intelligence, began. These applications are utilized to connect with customers in online, stores, such as through client support, commercializing, the amusement business, information gathering, and hybrid threats applied to influence public opinion (Dahiya, 2017).

**Eliza-** Eliza, a program developed by the MIT Artificial Intelligence Laboratory between 1964 and 1966, is one of the oldest and most well-known chatbots. Professor Joseph Weizenbaum created this program, which served as an inspiration to many other developers in the industry. Eliza is a character from G. B. Shaw's 1912 comedy Pygmalion, and the program is named for her. Eliza Doolittle, a lowly English street flower girl, comes to understand how to talk in a ladylike manner in this satirically critical piece, and her performance eventually impresses London high society. In the first circumstance, DOCTOR, Eliza chatbot plays the part of a psychotherapist, asking inquiries that she also replies to, diverting focus away from herself and toward the user. People began to anthropomorphize Eliza and find comfort in her and spread their tales, important info, and confessions, which came as a surprise (ZEMČÍK, 2019).

### 1.3 Problem Statement

As chatbots aren't human, they can't interact with patients as humans or doctors can. They sound overly mechanical, and they can only solve problems that have been programmed into them. They cannot respond to a patient of the circumstances, and they cannot display any emotions if necessary. Chatbots are also incapable of having a natural-sounding in-depth conversation with patients to understand their problems or disease, so they are only suitable for answering basic questions. However, this may cause a disconnect with users who prefer a human-to-human approach to problem-solving. As a result, they can only react to basic queries from patients and provide general information which they also possess accessibility to. They can't deal with complicated topics or answer queries that aren't on the script. Meanwhile, as time goes on, this is improving, and more powerful chatbots are now available. Natural Language Processing, which is increasingly popular for medical support, disease prediction, and suggesting a suitable doctor or hospital, is used to develop chatbots. Natural Language Processing is a branch of Machine Learning that allows users to communicate with machines through text and respond to their questions. However, Nowadays, to start good life, healthcare is most important. But sometimes, it is difficult for the patient to find a specific hospital or doctor for their disease. Sometimes it is also challenging to get an appointment with a particular doctor. It is not possible to visit the doctors or experts when immediately needed. In that emergency case patient can ask the chatbot about an appointment with the doctor. With web services in healthcare, education, and software solutions, chatbots are now more significant portals to online support and facilities. Although, there is a scarcity of information about chatbots' influence on consumers, organizations, and communities. Furthermore, before the full potential of chatbots can be achieved, several difficulties must be overcome. As a result, chatbots have become a popular research topic. We suggest a study plan in the form of future directions and issues to be addressed by the chatbot research to assist increase knowledge in this new research area.

## 1.4 Research Question

The question can arise that does this system can solve the existing problem?

Chatbots can solve a broad array of simple and complex problems in healthcare services.

Because they can ensure that-

- Assisting with Continued Lead Generation
- Guiding Users to Better Outcomes
- First Line users Service Support
- Increasing Productivity
- Feedback Loop and Suggestion Collection

## 1.5 Objective

This study aims to make a chatbot more efficient and helpful for the diabetes patients in Bangladesh. In this study, we will propose an advanced and brilliant message analyzing system for an intelligent chatbot, which will make the chatbot more intelligent and intelligent by using a natural language processing system.

The chatbot can simulate any procedure efficiently in our proposed message analysis by analyzing patients' messages. It can reduce any task completion time.

Objectives of this study are given below-

- The chatbot system will suggest nearest diabetes centers, hospitals and doctors for the diabetes patients in Bangladesh. At first the chatbot will ask or the patient will tell his home address or location to the chatbot and the chatbot will suggest nearest diabetes doctors and hospitals to the patients with their details.

- The intelligent chatbot will also provide a proper diet plan to the individual diabetes patients according to their blood glucose level and blood pressure level. Chatbot will also provide necessary information about diabetes to the patients.

- This chatbot system will let to know the diabetes patients that how is their diabetes conditions and situations into their body and exactly what they need to do or how they can diagnosis it. By all these methods the chatbot will help the diabetes patients in order to control their diabetes properly and to diagnosis the diabetes as well.

## 1.6 Contribution of this study

In this study, we will implement the natural language processing to implement the intelligent chatbot system. We will implement the chatbot system for diabetes hospitals or health care systems. Though some chatbots are available nowadays, they have huge limitations and cannot help people with the correct information or in a good way. We will implement a more advanced intelligent chatbot. Thus, these chatbots can help the users and the people with proper information. Also, therefore, these chatbots can chat with patients more accurately and fluently to provide them with the best service.

Chatbot apps increase the customer satisfaction by facilitating interactions between individuals and businesses. Simultaneously, they can provide clinics in Bangladesh with new chances to improve the quality of care. Many service centers employ people to provide solutions for ordinary citizens. If the chatbot can provide necessary assistance to diabetes patients, this workforce can be utilized in other fields. In this case, a chatbot used in a diabetes hospital system can understand the patient's queries. Analyzing their messages can provide a faster, less time-consuming, and more efficient solution.

### 1.7 Summery

This chapter shows the basic idea and introduces our proposed natural language processing system, which is an intelligent chatbot system for diabetes patients in Bangladesh using natural language processing. The problem statement section discusses the importance of our proposed advanced and intelligent chatbot system and, if the chatbot system is not available, what will be the disadvantages for that. There are also some critical research questions: Does the message analyzing system of intelligent chatbot solve people's problems in the diabetes hospitals by helping patients by providing proper patient care and giving appropriate hospital information service. Then in this chapter, the significance and contribution of this study also came out. It shows the advantages and the great importance of chatbot systems for diabetes patients in Bangladesh.

# CHAPTER 2

# LITERATURE REVIEW

## 2.1 Overview

This chapter describes the literature review, which shows the reviews on Message Analysis for the Intelligent chatbot system for providing medical support in Bangladesh using Natural Language Processing. Here we apply the chatbot for medical purposes and emergency medical support for the patients in Bangladesh. This chatbot also suggests particular hospitals and doctors for particular patients. Here also shows previous research and studies on chatbots and on implementing chatbots by using natural language processing. A discussion of alternative conceptual models of a chatbot for medical healthcare support, sentimental analysis methods, natural language processing methods, and past findings of these approaches and methodologies follows.

## 2.2 Natural Language Processing

Natural language processing (NLP) is preferable to computer science and artificial intelligence branches (AI). This enables computers to interpret text and spoken language in the same manner that humans can. At first, the natural language processing (NLP) concept came from the concept of machine translation. The idea of natural language processing came from the need for machine translation in 1940. At that time, the original language was only English and Russian. However, the use of other languages also came in the year 1960. Then in the year of 1964, the first chatbot was developed in the world, which was called "ELIZA". It was developed in the artificial intelligence laboratory of MIT, and it was developed by Joseph Weizenbaum (Sethunya R Joseph, 2016). It was a psychometric-based chatbot capable of taking the Turing test at that time.

Nowadays, there are much more use cases and works of natural language processing (NLP), and those use cases or works are discussed below.

- **Spam detection**

  Nowadays, people are facing spam messages or emails in their messenger or personal email. For detecting those spam emails and messages, NLP is used in order to recognize the fake mail or the spam mail. By using NLP, the emails and messages can be classified into which are more critical, which are less critical, and which are spam for the people. (Khurana, Koli, Khatter, & Singh, 2017)

- **Machine Translation**

  With the help of NLP/MT, people can now easily translate one language to another language. These tools are developed by using NLP, and these tools are capable of translating a word or language to another word or language very accurately. Now Google Translate is a significant example of it. (Khurana, Koli, Khatter, & Singh, 2017)

- **Virtual agents and chatbots**

  Speech recognition and natural language generation are used by virtual agents or chatbots to spot patterns in voice requests and respond with suitable, helpful comments. Nowadays, chatbots are developed using natural language processing, and these chatbots are capable of chatting with humans, almost similar to humans do chat with each other. Chatbots are now used in some medical centers and mainly on e-commerce websites. Chatbots analyze the messages of the customers or the patients and give an appropriate reply or suggestions in order to give a customer care or health care service. (Khurana, Koli, Khatter, & Singh, 2017)

- **Information Extraction from Sentiment analysis on social networking sites**

  By using natural language processing, story used in social media postings, answers, reviews, and other places may be analyzed using sentiment analysis. Here, analyzing any accident's data prediction of accidents is also possible now. (Chawdhary, 2020)

- **Text summarization**

By using NLP techniques, text summarization can be used in summaries. (Chawdhary, 2020)

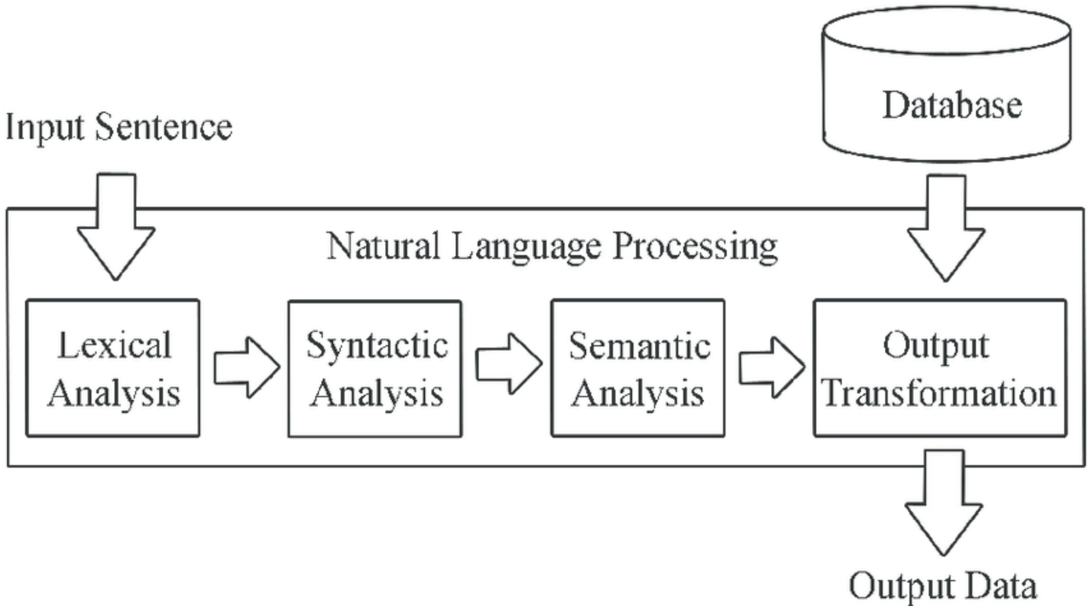


**Fig 2.1: Natural Language Processing Steps**

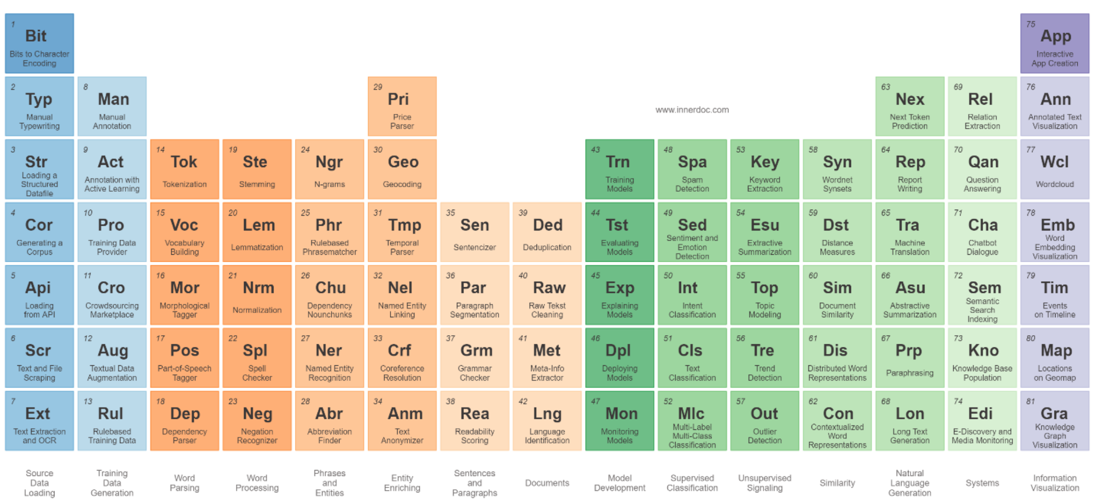


**Fig. 2.2:** Periodic Table of Natural Language Processing Tasks

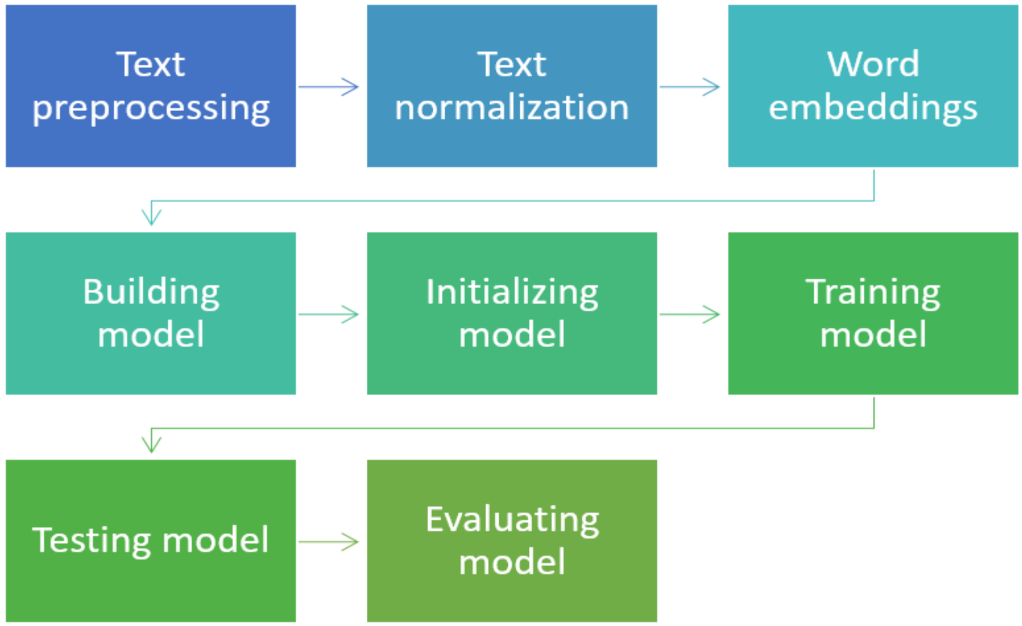


**Fig 2.3:** Natural Language Processing Tasks

### 2.3 Machine Learning

It is a global forum for research on computational learning methodologies (Murphy, 2015). It also enables software programs to improve their prediction accuracy without being specifically coded to do so. Machine learning includes natural language processing, and it strives to build a machine that understands or responds to text or voice data. There are some methods of machine learning. They are:

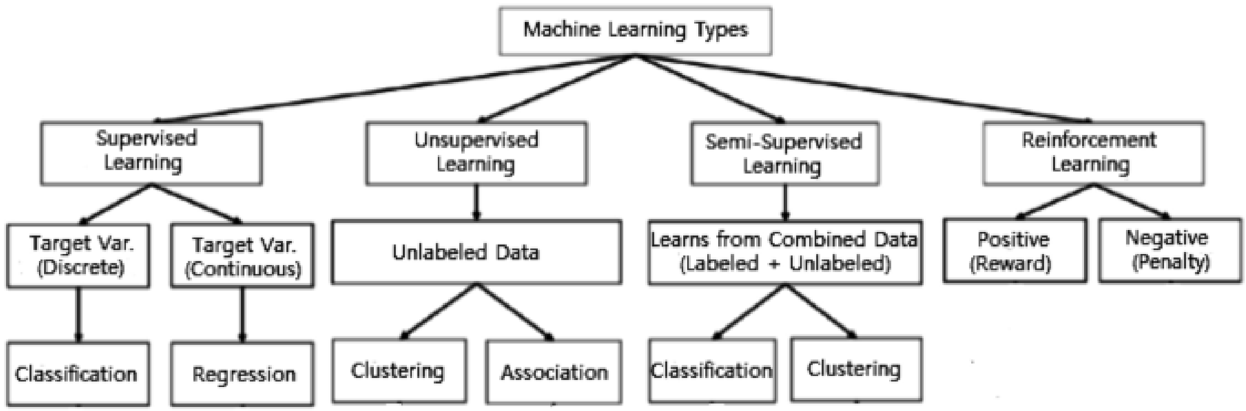


**Fig 2.4:** Methods and Classifications of Machine Learning

### 2.4 Chatbot

A chatbot is a software system that interacts with users that enables computers to talk and converse like humans. Initially, a chatbot can only respond to conventional inquiries whose questions and responses have been pre-programmed into the system. (Nuruzzaman & Hussain, 2018) With technical advancements, machines can eventually answer a freelanced question like a human by passing the Turing Test, which is a test of intellect that is closer to human intelligence. With the rapid advancement of artificial intelligence in recent years, intelligent chatbots have entered a new era and are now widely used in a variety of businesses. The computer performs Natural Language Processing (NLP) to overcome this obstacle. It is the process of identifying the input as well as the correct meaning of the specific action or need because of the occurrence of unknown and unexpected components in the input and the need to determine the appropriate understanding to apply to it. Intelligent chatbots apply machine learning to gain knowledge from the demands and details of the clients. Intelligent chatbots have been programmed to recognize words and phrases that are particular and that prompt a response. They gradually taught themselves to grasp additional queries and provide improving responses.

The concept of chatbot first came in the year 1964 when the first chatbot in the world was developed, and the name of this chatbot was "ELIZA". It was a psychometric-based chatbot. Eliza offers prefabricated responses utilizing "pattern matching" and replacement methods, making early users feel like communicating with real humans who understood their input. The program was limited by the scripts in the program (ZEMČÍK, 2019).

There are different versions of chatbot and those are:

- **Menu/button-based chatbots**

These chatbots are identical to interactive voice response menus in that they demand the user to make many choices in order to get to the final response. (Gupta & Hathwar, 2020)

- **Rule-based chatbots**

This version of the chatbot creates conversational flows using if/then logic. A rule-based chatbot is a chatbot that applies human-made rules to store, sort, and manipulate data. It copies human intelligence. When faced with a large number of similar inquiries, these chatbots fall short. If the keyword recognition fails or the user needs help finding an answer, these chatbots give users the choice of directly asking their questions or using the menu buttons on the chatbot. (Gupta & Hathwar, 2020)

- **Keyword recognition-based chatbots**

These chatbots use variant keywords and an AI program that uses natural language processing (NLP) to provide consumers with the most appropriate response or reply. These chatbots are known as NLP chatbots, and they give users the option of directly asking their inquiries or using the chatbot's menu if they need some help finding an answer. (Gupta & Hathwar, 2020)

- **Machine learning chatbots**

Machine learning chatbots are also called contextual chatbots and these chatbots are more advanced than any other types of chatbots. These versions of chatbots utilize machine learning (ML) and artificial intelligence (AI) in order to memorize about discussions with humans to improve. These chatbots are intelligent enough to improve themselves. The more they make conversations with the people, the more they can teach themselves. (Nuruzzaman & Hussain, 2018)

### 2.5 Chatbot related recent works

Nowadays, chatbots are very popular in the world. In this era, chatbots are created mainly by using python programming language. Other advanced programming languages are also used for creating chatbots like C# and Java programming language. The college is called St John College of Engineering and Management, Palghar, in India. There was research on AI-based health care chatbot systems in that college using natural language processing. The college students did the research. The research has the purpose to make a healthcare chatbot based on Artificial Intelligence using NLP to detect and diagnose a specific disease and provide required details about the specific disease before visiting a doctor. The purpose of the research is also to reduce healthcare costs and improves accessibility to the medical chatbot. They made a chatbot that acts as virtual medical assistance, which helps the patients to know more about their disease (Harsh Mendapara, 2021). Another research paper on a chatbot called Aquabot, was published in the International Journal of Advanced Computer Science and Applications (IJACSA) in 2017. The research was done at SE Foundation University Islamabad, Islamabad, Pakistan. The purpose of the study was to create a chatbot that will be used to help Diabetic patients. Based on the user's text-based queries, the chatbot may identify the severity of an illness. It extracts meaning from text using natural language processing (NLP) and utilizes decision trees to classify a patient in terms of illness risk. (Sana Mujeeb, 2017). Another study was conducted to create a chatbot architecture based on natural language processing as well as machine learning providing public healthcare help. The research was held in List Laboratory, Faculty of Sciences and Techniques Bp416 Tangier Morocco. The goal of the study was to propose a practical architecture for creating an intelligent chatbot to assist with health care (Soufyane Ayanouz, 2020).

**Advantages of Chatbots**

- Chatbots can make the customer service or medical healthcare services available 24 hours every day.
- A chatbot can save the time and money of a user.
- Chatbots eliminate face-to-face communication with clients.
- A chatbot can eliminate painful, time-consuming tasks
- A chatbot can eliminate interactive voice response (IVR) systems
- It can monitor the health-related issues of a patient
- It can help patients by providing necessary medicines for a specific disease.

**Disadvantages of chatbots**

- Chatbots cannot interact as a human with customers. They sound too mechanical.
- Chatbots are only capable of answering simple queries.
- Chatbots are challenging to create
- Chatbots require constant maintenance

## 2.6 Summery

The literature review for the suggested chatbot system, a diagnosis chatbot system for diabetes patients in Bangladesh, was discussed in this chapter. This chapter covers natural language processing (NLP), machine learning (ML), and chatbot principles, types, and related works. Natural language processing and machine learning principles were discussed. Many types of test cases of NLP and many types of machine learning are currently used by people worldwide. Supervised learning, unsupervised learning, and semi-supervised learning and their working method are discussed briefly in this chapter. Concepts of a chatbot, history, and many types of chatbots used in e-commerce sites and for medical purposes are also discussed. The advantages and disadvantages of a chatbot system are also shown in this chapter.

# CHAPTER 3

# METHODOLOGY

## 3.1 Overview

This chapter discusses the research methodology and framework of our proposed system, an intelligent chatbot for diabetes patients using natural language processing in Bangladesh. We also discuss the feature-driven development (FDD) system method, which we will follow to develop our chatbot system. We will also show the total cost estimation and budget of our proposed system or project. This chapter will show the total amount of cost of developing this project. We will also discuss the project schedule, time estimation, and our proposed chatbot system management by creating a suitable timetable. In the table, we will show the entire working process of our proposed system with the total time estimation and schedule.

## 3.2 Research Framework

In our proposed intelligent chatbot system, the study proposes a new framework. This framework shows the working process of the chatbot system. In this framework, it can be seen that the conversation between the diabetic patient and the chatbot. In the conversation, according to the patient's questions, the chatbot will suggest nearby diabetes centers, doctors, and hospitals for the patients. The chatbot will also provide necessary information about diabetes to the patients.

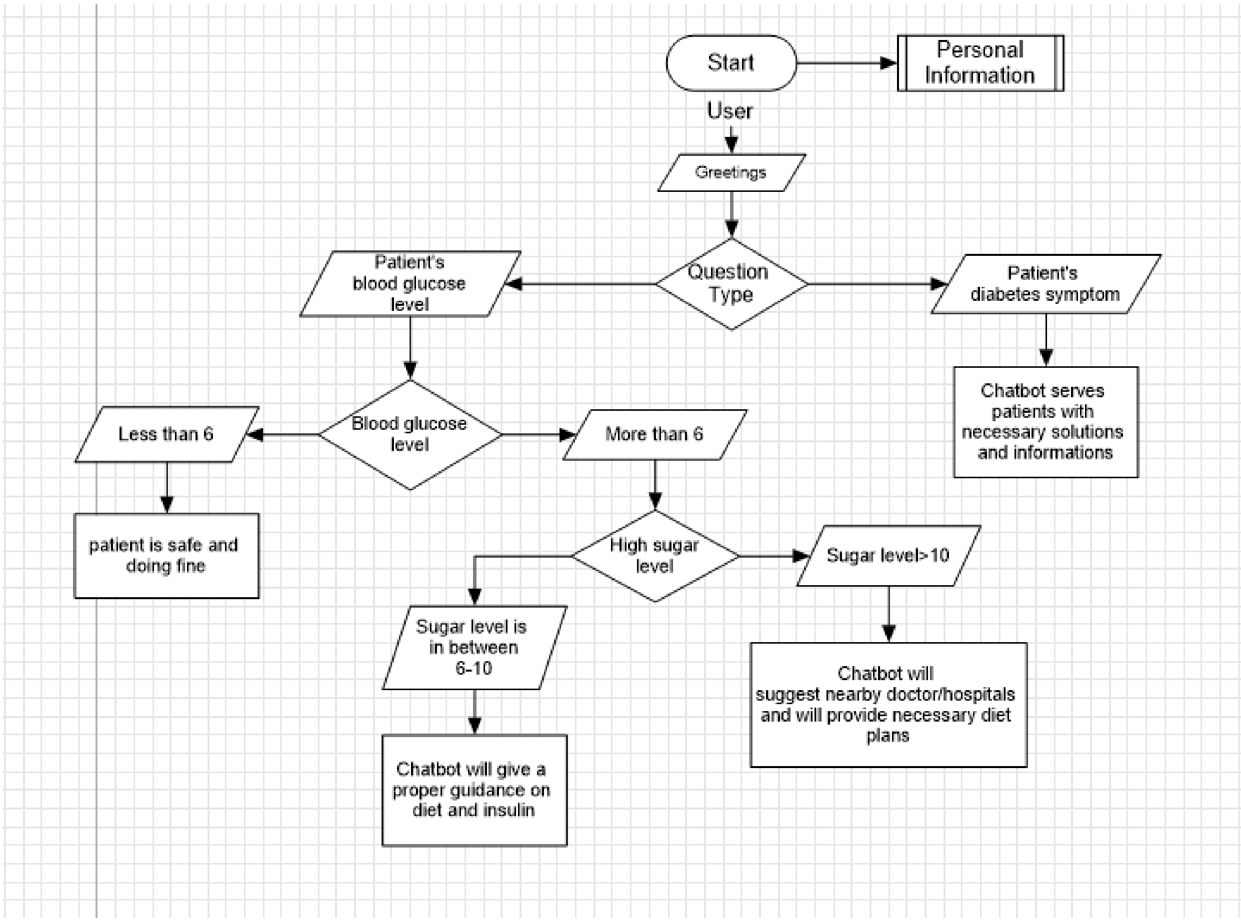


**Fig 3.1 Proposed Framework of Chatbot for Diabetes Patients**

After greetings with the chatbot, the patient can give his information about his blood glucose level and diabetes symptoms to the chatbot. According to the patient's queries and location, the chatbot will suggest him/her nearby diabetes doctor or specialist and diabetes hospital. The chatbot will also provide necessary diet plans to the patients. The chatbot will also provide all diabetes doctors' lists and details about the doctors.

In our proposed intelligent chatbot system, we will use the natural language processing (NLP) here in order to make the chatbot more intelligent, more fluent, and intelligent in chatting with the patients or users. The study of this chatbot using NLP proposes a new architecture framework that will define the uses of natural language processing in chatbot very clearly.

By using natural language processing, chatbots can use intents and sentimental analysis to infer and gather the appropriate data responses to deliver higher accuracy rates in the responses they provide. Using the NLP chatbot can have comprehensive communication with the patients and understand the patient's problems and symptoms faster. NLP retrieves symptomatic keywords based on user responses to assess the user's current medical problems. By this method, the user or patient will feel like they are having a dialogue with a health professional (Gopi Battineni, 2020). According to the symptoms and problems, a chatbot can provide the best and most straightforward solution for diabetes patients. Using statistical analytics and natural language processing, the chatbot can recognize user input problems and resolve difficulties.

### 3.3 Research Methodology

In this study, we will follow the feature-driven development method (FDD), which is an agile methodology. The feature-driven development approach (FDD) is an iterative and incremental strategy for developing the finest software or AI system, and it makes it easy to track progress and outcomes. In the FDD method, teams update the project regularly and can identify errors quickly. This method helps to reduce confusion and rework. Feature-driven development is best for a structured and focused Agile methodology, which can be scaled across the product organization and deliver clear outcomes. (Mahdavi-Hezave & Ramsin, 2015)

In 1997, Jeff De Luca first invented and discovered this feature-driven development method (FDD). In Singapore, he was part of a 50-person team working on a 15-month software development project. He designed a development strategy that includes five steps and focuses on producing features in short iterations to assist the development team in being more adaptable and responsive to customer requests. Jeff De Luca and Peter Coad worked on a project at United Overseas Bank in Singapore as part of a team. The Coad Method, which Peter Coad proposed, was widely recognized. The Coad Method was primarily concerned with the employment of object technology in large-scale line-of-business IT projects. The Coad Method developed into Feature Driven Development as additional fresh ideas were contributed across the research work in Singapore. (Huda, 2011).

Other agile methodologies Scrum, XP, and other agile approaches, all use a step-by-step procedure to software development. The five phases of FDD, on the other hand, require the team to adhere to a set of engineering best practices while developing tiny feature sets in one- to two-week iterations. These five processes ensure that development stays continuous, allowing the project to grow and new team members to catch up quickly.

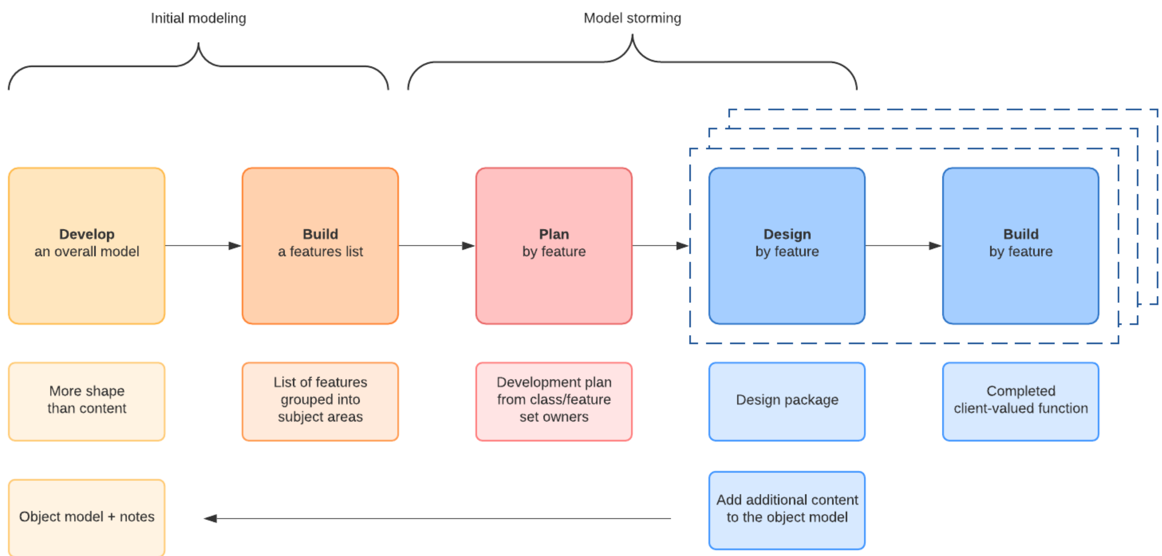


Fig 3.4 Feature Driven Development Method

Nowadays Scrum method is so much popular, but rather than scrum FDD is more valuable and better for an AI system development. In this study, the chatbot system is an AI system and a featured system as well. That is why for a feature-based system, FDD is most popular, and here the feature-driven development will be chosen for the intelligent chatbot system. Scrum, XP, and other agile approaches use a step-by-step approach to software development. The five phases of FDD, on the other hand, require the team to adhere to a set of engineering best practices while developing tiny feature sets in one- to two-week iterations. These five processes ensure that development stays continuous, allowing the project to grow and new team members to catch up quickly.

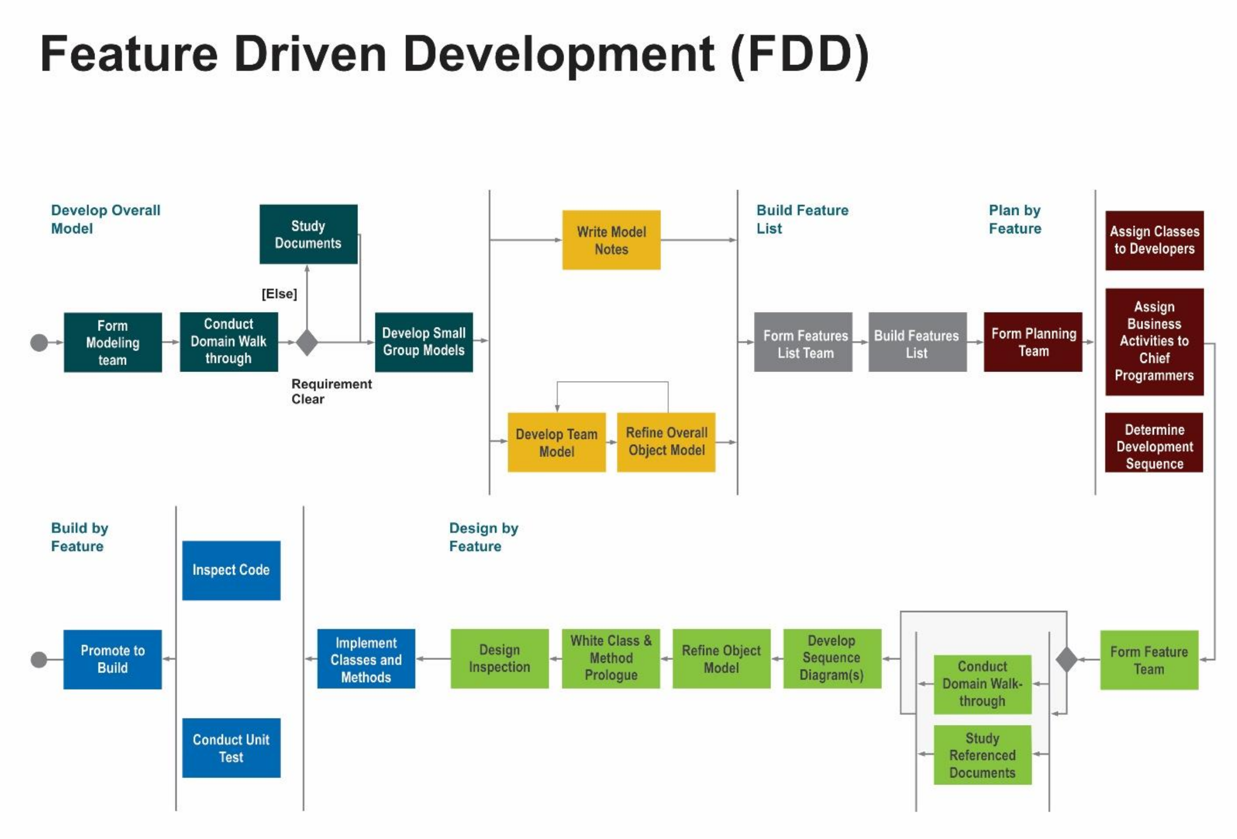


Fig 3.5 Working Process of Feature Driven Development (FDD)

The five primary activities of FDD are:

- Develop overall model
- Build feature list
- Plan by feature
- Design by feature
- Build by feature

### 3.4 Budget

In order to build this proposed intelligent chatbot system, a total amount of BDT 1,20,000 TK will be needed. A high-performance computer will be used for this proposed system.

| Components | Price |
|---|---|
| Processor:<br>intel core i7 7700K | BDT 19,000 |
| Motherboard:<br>MSI Z170A Gaming M7 | BDT, 20,000 |
| Ram:<br>Corsair Vengeance RGB pro 16GB(8x8) | BDT 8,000 |
| GPU:<br>Nvidia Geforce GTX 1070ti | BDT 45,000 |
| CPU Cooler:<br>Cryorig air cooler | BDT 4,000 |
| Case:<br>Corsair Spec Omega | BDT 9,000 |
| Power Supply:<br>Thermeltake smart BX1 RGB 750 watt | BDT 8,000 |
| SSD:<br>Western Digital 240GB | BDT 3,000 |
| HDD:<br>Seagate Baracuda 2TB | BDT 4,000 |
| | TOTAL: BDT 1,20,000 |

This Computer is used to build an intelligent chatbot system.

### 3.5 Project Schedule

In order to complete this research, two semesters, which means a total of 8 months was needed. A total project schedule estimation is shown below:

| Month(The year 2021-2022) | Project Work |
|---|---|
| September-October | Information gathering and researching on Chatbot System. |
| November-December | Introducing the diagnosis chatbot System for diabetes patients in Bangladesh. Researching on natural processing and the impact of NLP in the chatbot system or an AI system |
| January-February | Working on implementation By using Python programming and SQL database. Datasets, NLP, and machine learning methods are also used. |
| March-April | Testing chatbot and gathering feedback. Feature-driven development is used for this project. Gathering necessary information on diabetes and researching diabetes. Finally, the whole implementation was completed. |

### 3.6 Summery

This chapter it is discussed about the research methodology which is used to build the whole project or chatbot system and the method is an Agile method which is called Feature Driven Development (FDD) method. Here also discussed about the architecture proposed framework of our proposed diagnosis chatbot system. Natural language processing (NLP) is focused on here and the working process of NLP for a chatbot system is described clearly and in detail in this chapter. It is also discussed Why the FDD is necessary and essential for this project.

# CHAPTER 4

# IMPLEMENTATION AND EVALUATION

## 4.1 Overview

This chapter describes the implementation of the proposed model, which is the "Neural Network Model" for Intelligent Chatbot System for Diabetes Patients using Natural Language Processing in Bangladesh. This chapter also shows the evaluation of the proposed system, the complete implementation of the chatbot system, and the implementation of the “Neural Network Model” for the chatbot system. This chapter is divided into six parts; section 4.2 describes about the model and implementation of the proposed chatbot system. Section 4.3 describes the design of the chatbot. Section 4.4 describes about the training data of the model. Section 4.5 shows the result analysis; section 4.6 describes the findings and section 4.7 describes the evaluation of the whole process.

## 4.2 Development of the Model

The proposed intelligent chatbot system "Neural Network" model is used, and by this, neural network machines or chatbots are trained by some or many datasets. Trained datasets are given to the machine as input, and machines must learn to spot patterns in this data. After that, The results of a new batch of comparable data are predicted by machines. Artificial neural networks, or "Neural Networks," have been driven by the realization that the human brain computes entirely differently from a traditional digital computer since its conception. The brain is a parallel, nonlinear, and incredibly complicated computer (information-processing system). It has the ability to organize its structural parts, known as neurons, in such a way that it can execute specific computations (such as pattern recognition, perception, and motor control) many times faster than the current fastest digital computer. Consider human vision, which is an example of an information-processing task. The visual system's job is to portray the environment around us and, more importantly, to deliver the information we need to interact with it (Haykin, 2009). In the development of the chatbot system, bot "Neural Network and "Natural Language Processing" are used to acknowledge the text data of the user or patient and give an appropriate specific reply.

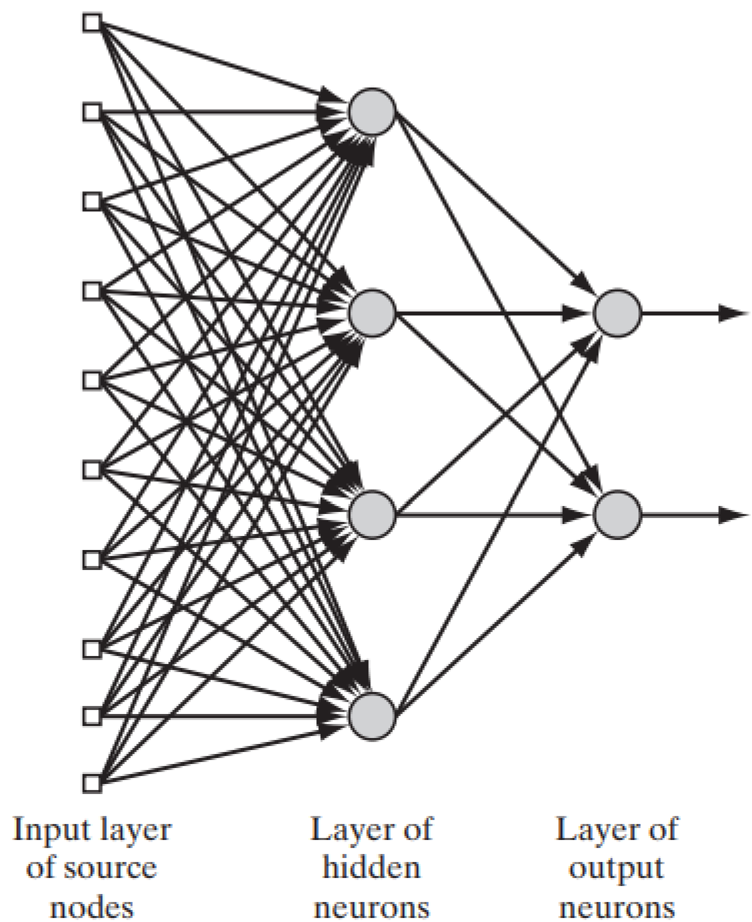


**Fig 4.1 Working Process of Neural Network Model**

### 4.2.1 Implementation

During the implementation of the proposed chatbot system, the deep learning method, which is the "Neural Network" method, and the "Natural Language Processing" method are both applied here. (Zhou, 2020) For the greetings with the chatbot, the bag of words (Zhang, Jin, & Zhou, 2010) is applied here in order to recognize and analyze the greetings of the patient or user like hi, hello, hey, how are you, etc.

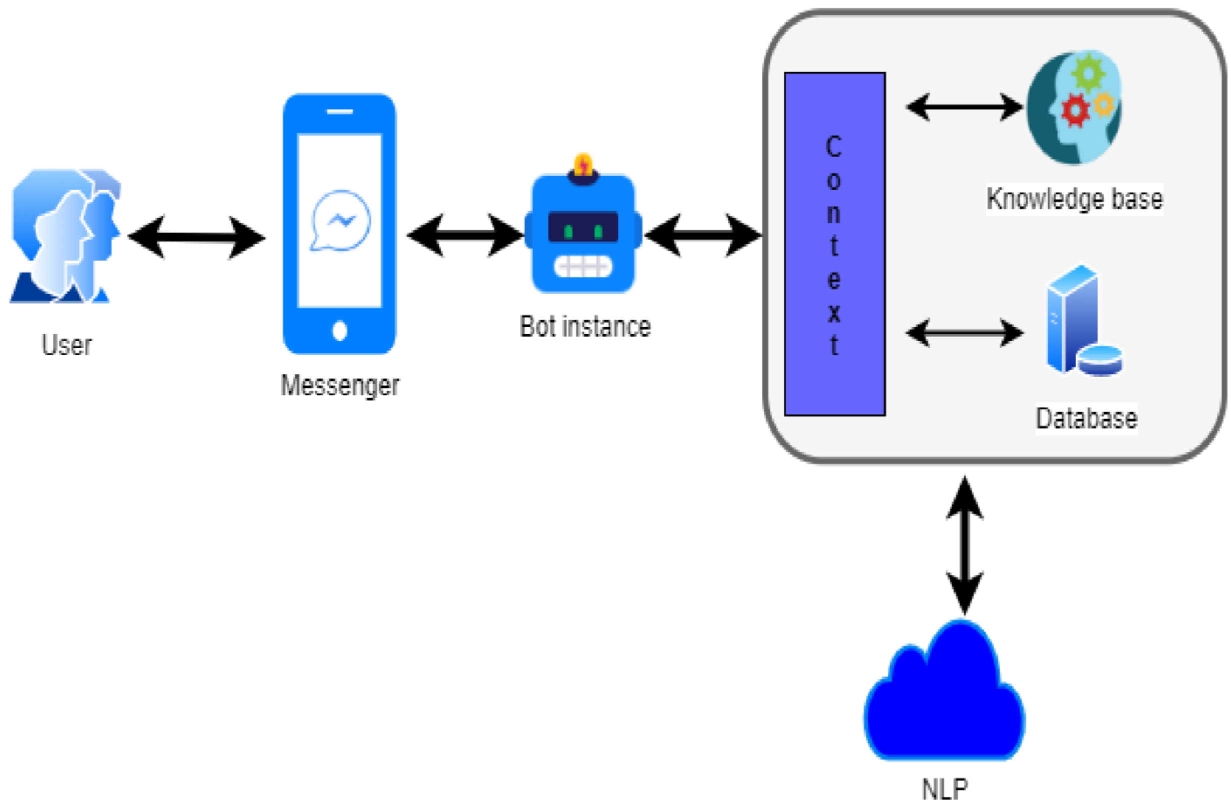


**Fig 4.2 Architecture of Chatbot's Functionality using Natural Language Processing**

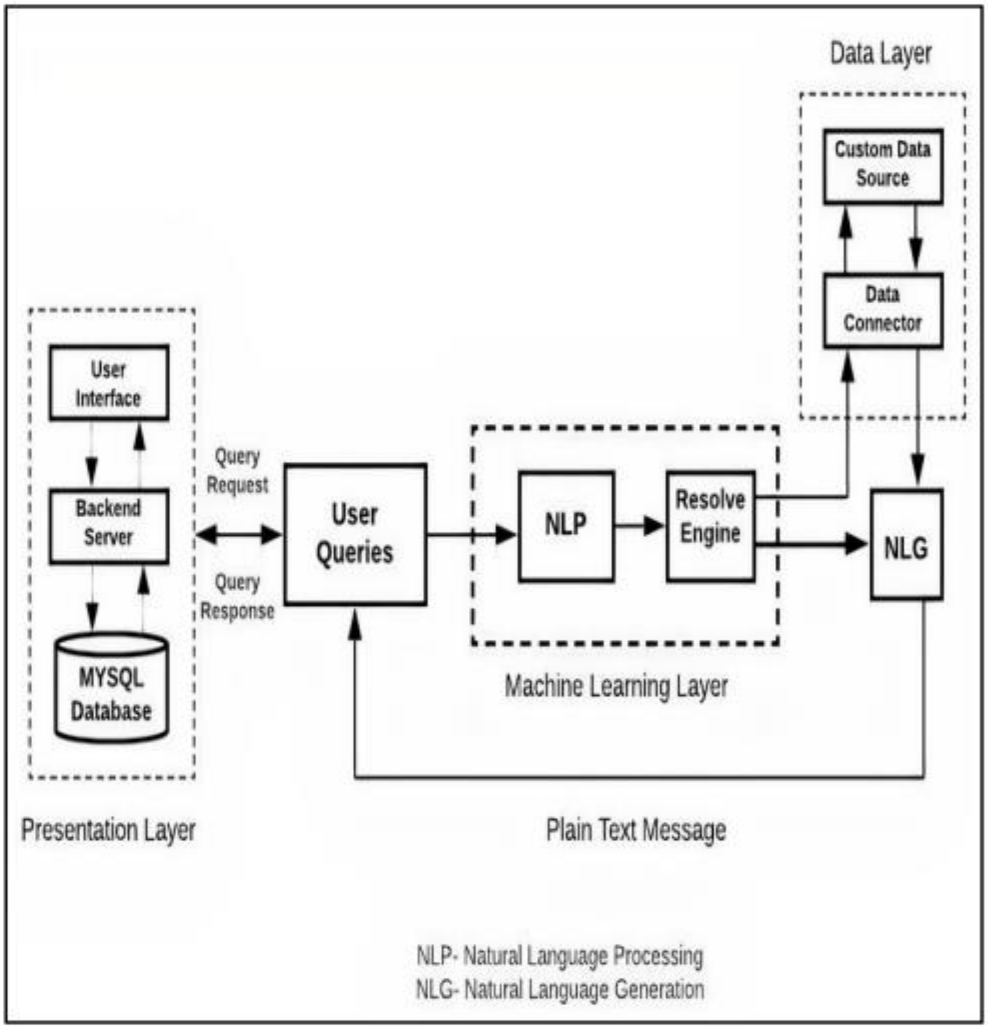


**Fig 4.3 Proposed System Architecture of The Intelligent Chatbot for Diabetes Patients**

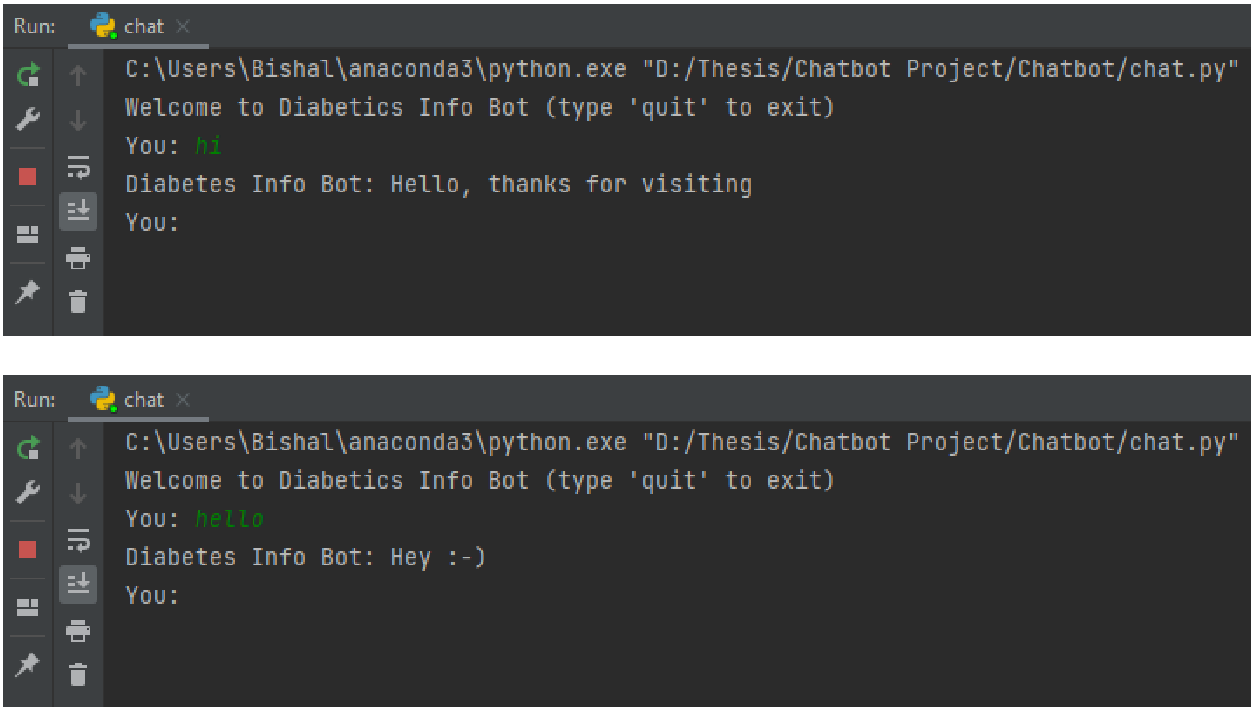


**Fig 4.4 Greeting System with The Chatbot**

Next, the Tokenization is used here also a natural language toolkit (NLTK) to break down the text data in the system to recognize and make the machine understand precisely what the patient or the user is asking or saying. By tokenizing a sentence into the text dataset, a machine or system can understand and recognize the words individually in the sentence and can classify them in an appropriate way or method.

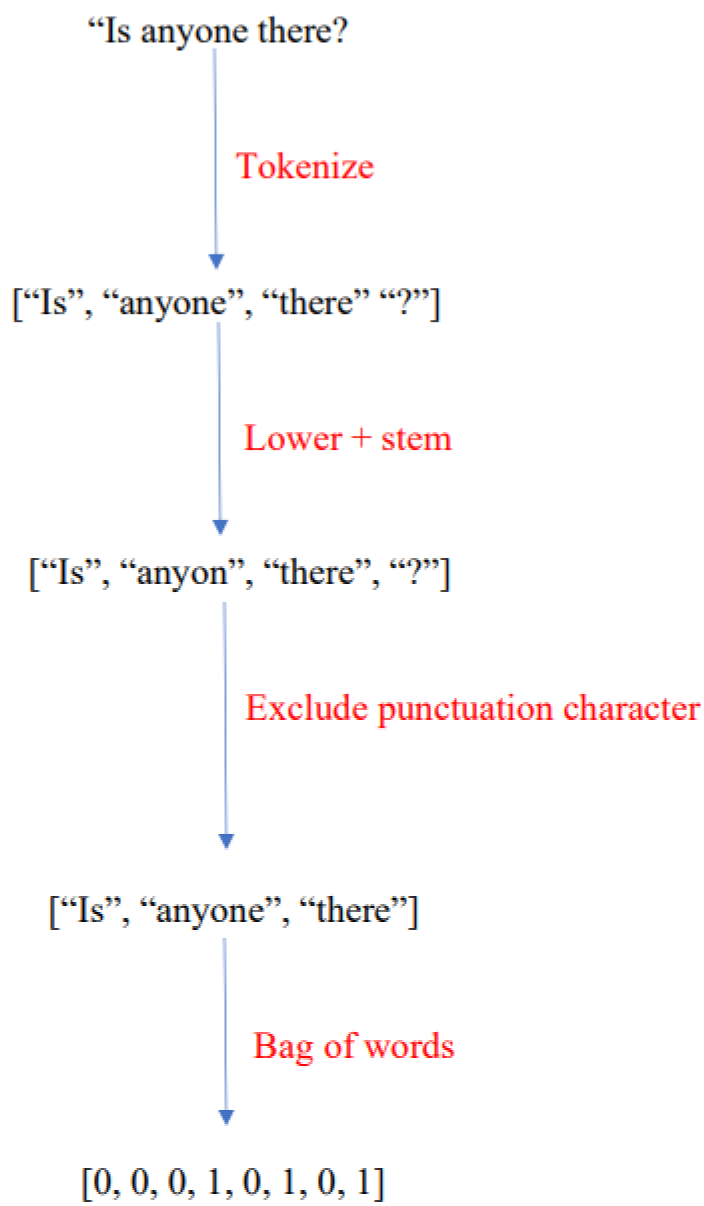


**Fig 4.5 NLP Preprocessing Pipeline**

Then by using the stemming (NLTK took), the words are reduced into their word stem in the text datasets. This natural language system is used here to recognize, search and retrieve more forms of words which returns more results, and by stemming, the chatbot system finds the root words of the words in the sentences of the test dataset. Then by using the "Neural Network" model, the conversation system with the chatbot is created here. The "Feed Forward Neural Network" is used in the proposed chatbot system. This type of neural network is used in the proposed chatbot system in order to understand the human languages and classify the languages in the machine to train the chatbot system. Thus the system can understand the human language and can give an appropriate output or reply. (Sazli, 2006)

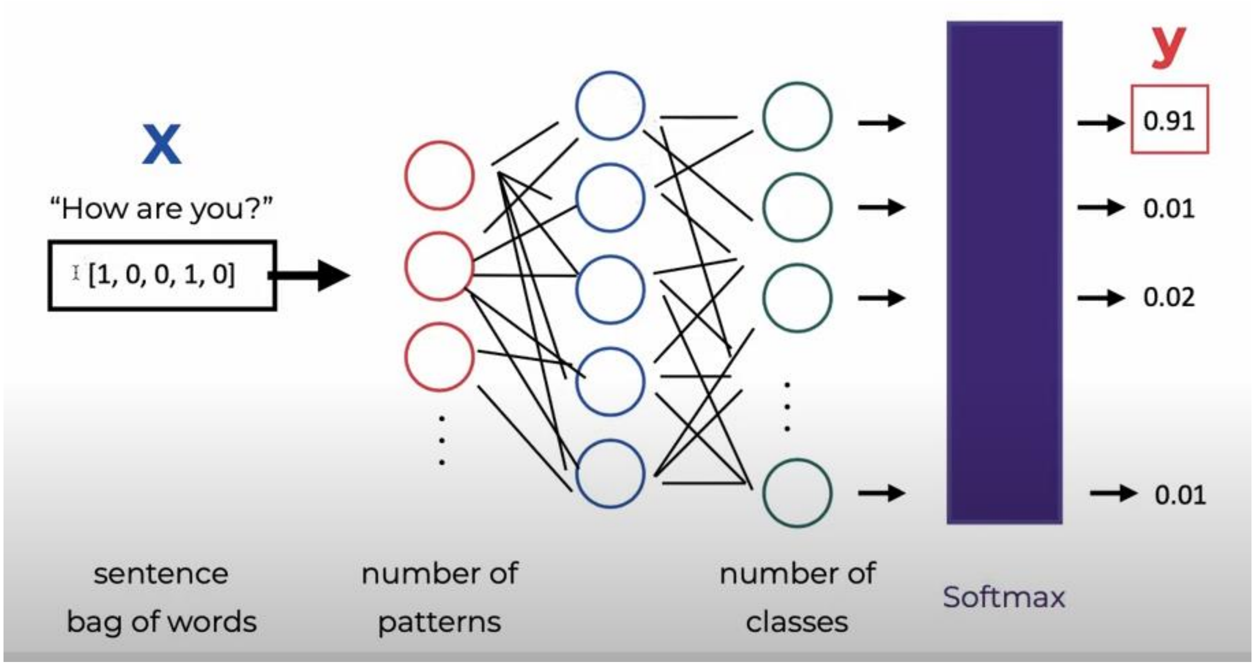


**Fig 4.6 Feed Forward Neural Network**

In fig 4.4, we can see the architecture and the working process of the "Feed Forward Neural Network" model. The bag of words is given as input in the figure, and the number of patterns is the input size here. Then the input patterns are connected with the hidden layers, and from the hidden layers, it goes to the output size, which is the number of different classes. Then the softmax is applied here in order to get the probabilities of different classes.

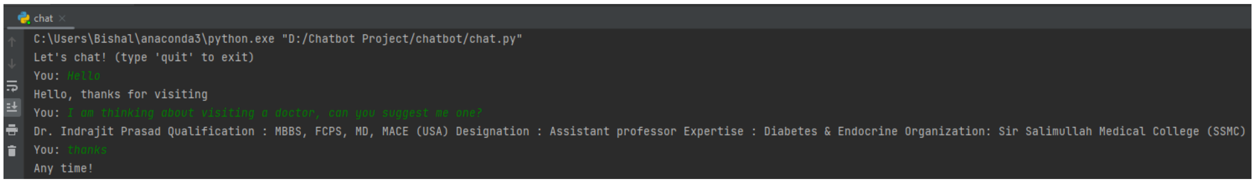


**Fig 4.7 Conversation system with the chatbot**

### 4.2.2 Implementation Tool

By using the "Neural Network" Model and Py Torch tool, the chatbot is implemented in the PyCharm compiler via python programming. In order to use natural language processing here, we use Natural Language Tool Kit (NLTK), and those tools are

- **Bag of Words**

  In this chatbot system, we use the bag of words, which is a natural language processing technique of text modeling, to build structured, well-defined fixed-length inputs. By using the Bag-of-Words technique, the variable-length texts are converted into a fixed-length in the test dataset. Here the text datasets are converted into an equivalent vector of numbers.

- **Tokenization**

  In the implementation of this intelligent chatbot system, we are also implementing Tokenization, and when it comes to working with text data, this is one of the most common tasks.

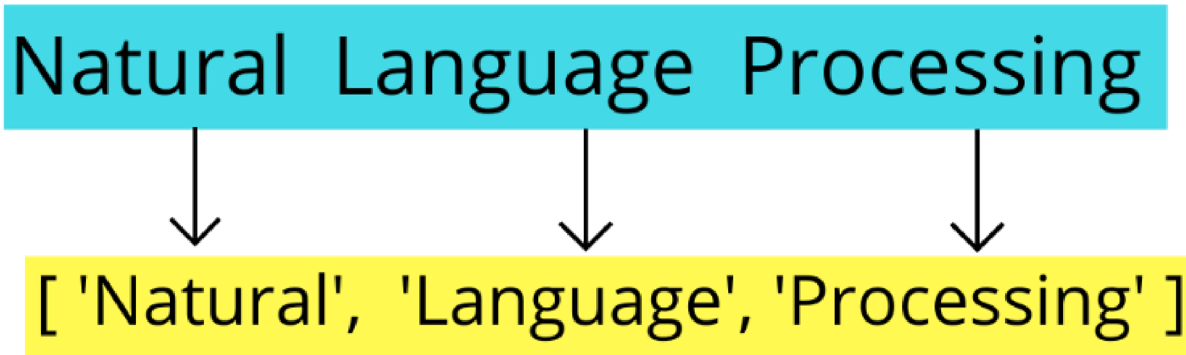


**Fig 4.8 Tokenization Technique in Natural Language Processing**

- **Steaming**

  Steaming is a natural language processing technique used in the chatbot system to inflect words into their root forms, hence aiding in preprocessing text, words, and documents for text normalization in the test dataset. Having numerous inflected forms inside the exact text adds redundancy to the NLP process, even though a word may exist in several inflected forms. As a result, we use stemming in the chatbot system to break down words into their simplest form, or stem, which may or may not be a valid word in the language. As example:

Organize → Organ
Organizing → Organ
Organizes → Organ

Adjustable → Adjust
Formality → Formal
Airliner → Airline

### 4.3 Designing of Chatbot

In the design section of the chatbot system, the chatbot system is created in the PyCharm community edition compiler, and the PyTorch machine learning framework is used in order to create the environment of the chatbot and the chatbot as well. After developing the chatbot with PyTorch for testing purposes on the other devices, the chatbot system is deployed in the "ngrok," which is a programmable network edge that adds connectivity, security, and observability with an app with no code changes. Using the "Flask" python web framework, the chatbot is connected with the "ngrok." Then the chatbot is designed and connected with the PyCharm.

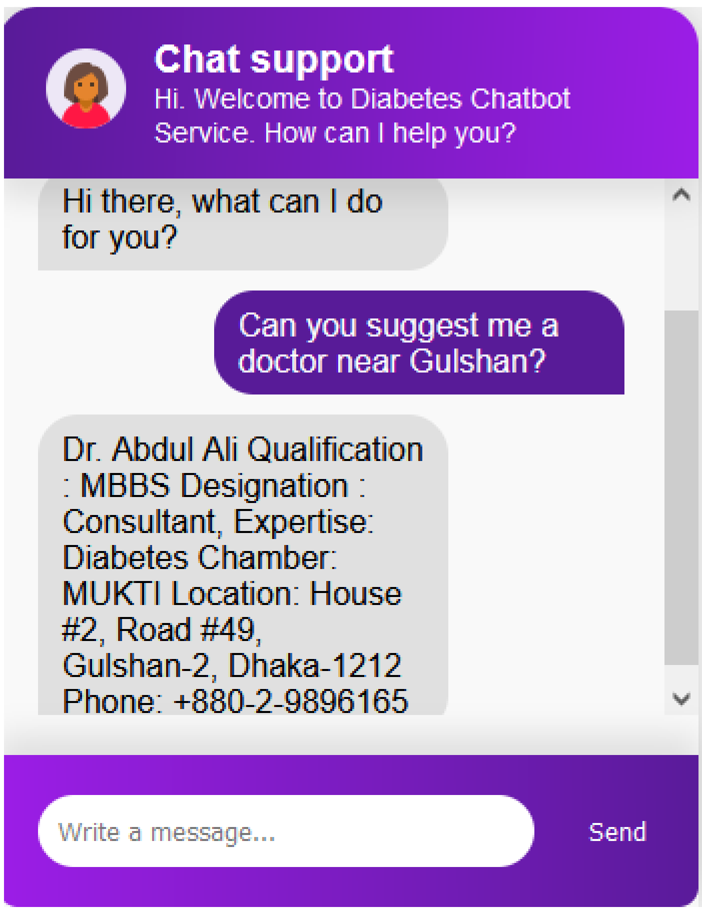


**Fig 4.9 Design of the Chatbot System**

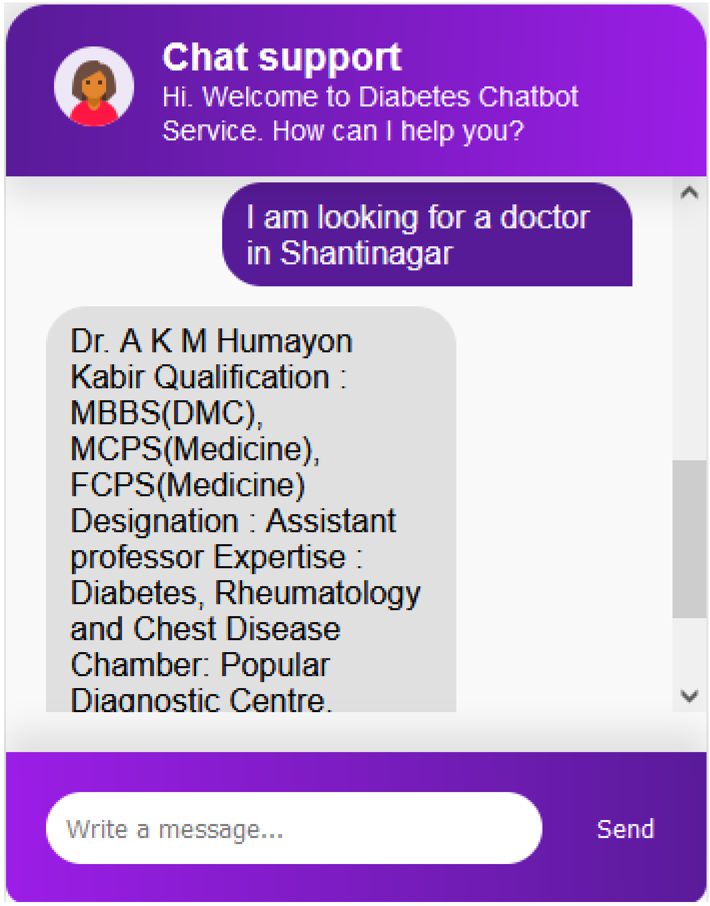


**Fig 4.10 Design of the Chatbot System**

### 4.4 Evaluation

This project includes an evaluation procedure in order to test the chatbot system and measure its effectiveness and efficiency of the chatbot. The primary purpose of evaluating the chatbot system for the diabetes patient is to check the accuracy level of the conversation with a patient and check the ability to give correct answers to the user's or patient's questions. Another purpose is to make sure whether the chatbot system is effective in Bangladesh or not.

### 4.5 Result and Analysis

During the testing period, some ten trained questions are given to the users and patients to test whether the chatbot can give the correct answers to the given questions. Among the users 5 of them are university students, 3 of them are middle aged people, 2 of them are diabetes doctors and 5 of them are diabetes patients.

From 15 users, the testing survey is taken here, and the output or result of correct answers given the ability of the chatbot system is 100% here. The survey also shows here that 66.7% of the users are satisfied with using this chatbot system for diabetes patients. They strongly agree that this chatbot system will be enormously effective for diabetes patients in Bangladesh. About 26.7% of users say that it will be effective for diabetes patients in Bangladesh, and about 6.6% of users say that it will be less effective for diabetes patients in Bangladesh.

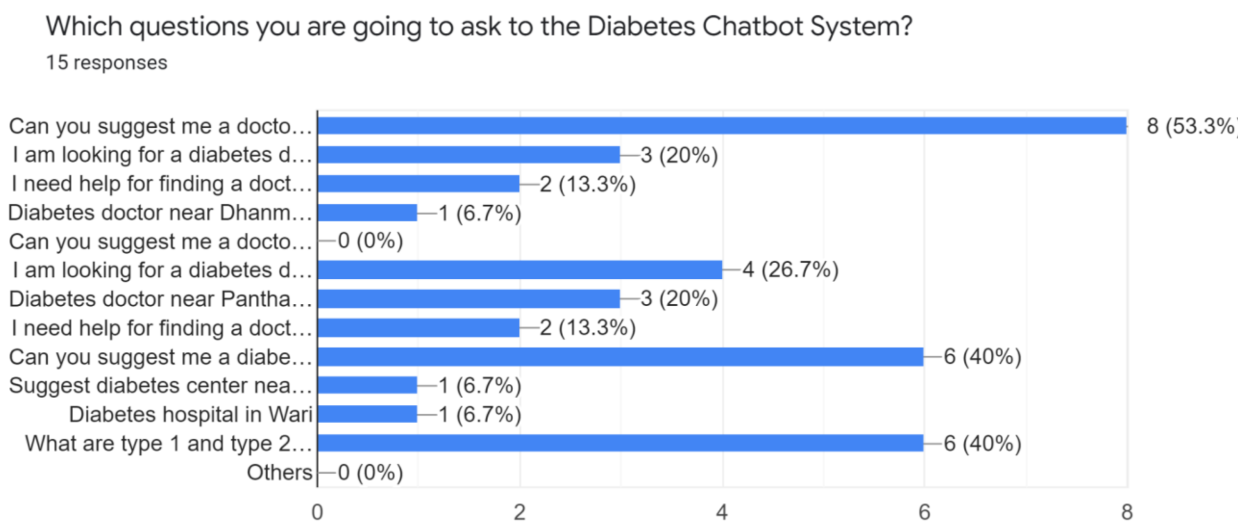


Fig 4.11 Ratio of Asked Questions to the Diabetes Chatbot System

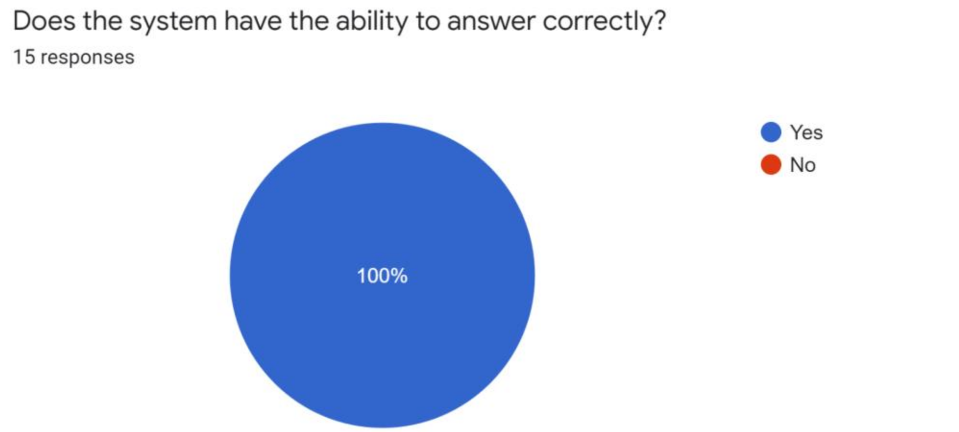


Fig 4.10 Result of Correct Answer Giving Ability of the Chatbot System

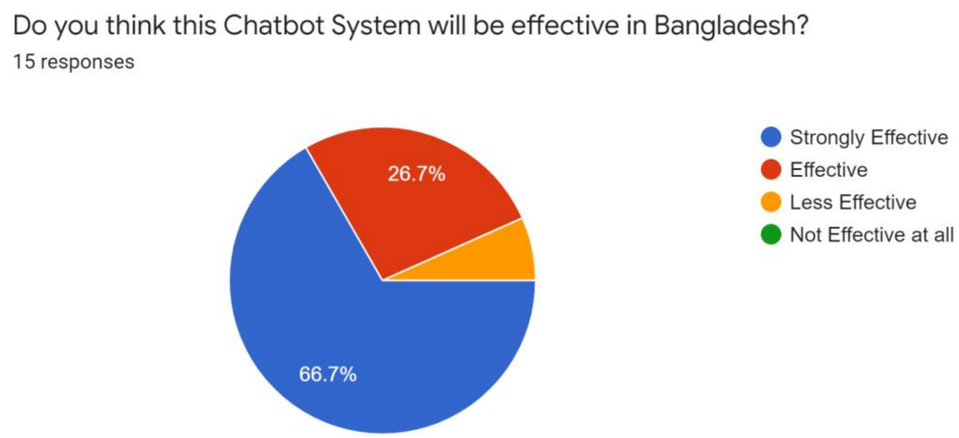


Fig 4.12 Result Analysis of the Chatbot System For Diabetes Patients In Bangladesh

## 4.6 Findings

In this study, an intelligent chatbot is created for diabetes patients in Bangladesh by using the deep learning method "Neural Network" and the use of "Natural Language Processing". The purpose of the chatbot system is to suggest the nearest doctors and hospitals for diabetes patients in

Bangladesh, and the chatbot system will also provide necessary and adequate information about diabetes. In this study, the chatbot system is also trained to give diet planning and necessary diet suggestions to diabetes patients. Thus, diabetes patients can get proper help, and diagnosis for the top-level disease in Bangladesh called diabetes.

# CHAPTER 5

# CONCLUSION

## 5.1 Overview

The core of the study is described in this chapter, which is based on the study's key objectives and the overall result and evaluation analysis. This chapter summarizes all of the study's work and findings. This chapter also summarizes the study's design, implementation, and significance, which is the chatbot system for diabetes patients in Bangladesh. This chapter also explains the limitation and future works of the proposed chatbot system.

## 5.2 Summary of Background

Nowadays, machine learning is prevalent, and it uses algorithms and neural network models to help computers improve their performance over time. Machine learning is instrumental and famous in the medical and healthcare sectors in this era. Among many other health-related issues, diabetes is a significant health-related issue and a significant disease in Bangladesh. "Natural Language Processing" is a most significant part of machine learning. It's a type of artificial intelligence that allows computers to understand text and spoken language in a similar way as humans can. Using "Natural Language Processing" and the "Neural Network" model, many chatbots were created previously for business and health-related purposes.

## 5.3 Summary of Objective

The primary purpose and goal of this study is to create a chatbot for diabetes patients in Bangladesh with the support of "Natural Language Processing" in order to suggest the diabetes centers, hospitals and doctors. Also, in order to provide necessary diet plans and information services to diabetes patients in Bangladesh. The chatbot can chat with the patients very frequently and precisely. Thus, it can provide the necessary services by message analysis system using Natural Language Processing (NLP).

## 5.4 Summary of Methodology

The model is used here, the "Neural Network" model, a Natural Language Processing (NLP) model, to create the chatbot system for diabetes patients in Bangladesh. This model is implemented by the Python programming language and PyTorch, which is an open-source machine learning framework. Bag of words, tokenization, and stemming of these three natural language toolkits (NLTK) is used here to analyze messages very precisely and correctly in the chatbot system. The "Feed Forward Neural Network" is a type of neural network model which is used here to build the whole environment and communication system of the chatbot system.

## 5.5 Summary of Design and Implementation

For the design of the chatbot system, the chatbot system is created in the PyCharm community edition compiler, and the PyTorch machine learning framework is used in order to create the environment of the chatbot. Later, the chatbot system is designed and connected by using " ngrok " a programmable network edge. By using the "Flask" python web framework, the chatbot is connected with the "ngrok". In the implementation part, the chatbot system is implemented by using natural language processing (NLP), neural network model and natural language toolkit (NLTK).

### 5.6 Summary of Evaluation

In order to test the chatbot system, an evaluation is done here. In the evaluation section, it can be seen that there are ten random questions asked by the users and patients to the chatbot, and the result is that the chatbot can give all of the question-answer correctly. The percentage of deploying the correct answer of the chatbot was 100% which was an outstanding and excellent result. After using the chatbot, a question was asked to the user and the patients what did they think the chatbot system would be effective in Bangladesh or not. 66.7% of users said it would be enormously influential, 26.7% of users said it would be effective, and 6.6% of users said it would be less effective.

### 5.7 Summary of Findings

After the development and implementation of the chatbot, the demo version of the chatbot system was very well, and the users' feedback was about 66.7% strongly effective which was an excellent result. As the survey result says, this chatbot will be very effective and useful for diabetes patients in Bangladesh. The chatbot's message analyzing system is very accurate because the "Natural Language Processing" method and the "neural network" model are used to develop the chatbot.

### 5.8 Significant of the study

In this study, an intelligent chatbot system is created for diabetes patients in Bangladesh in order to suggest to the patients nearby diabetes centers, hospitals, and doctors for them. The chatbot system will also provide the necessary information and diet plans to diabetes patients in

Bangladesh. By using the natural language processing (NLP) and neural network model, the chatbot's environment, design, and model are created. Nowadays, a chatbot is a beneficial and essential asset in this modern era. The chatbot can precisely understand what the users want or look for by the message analyzing method. After understanding, chatbots can reply accurately by the use of the neural network, NLP, and machine learning. If the chatbot Can provide necessary services and solutions, then this workforce can be utilized in other fields. In this case, a chatbot used in a hospital system can understand the patient's symptoms and problems. Analyzing their messages can provide a faster, less time-consuming, and more efficient solution.

## 5.9 Limitations and Future Works

This proposed chatbot system is now used for suggesting nearby diabetes centers, hospitals, and specialists or doctors to diabetes patients in Bangladesh. This chatbot system has some limitations and many possible future works. After finishing the complete development of the chatbot and after adding all the other necessary features to this chatbot system, The future works will be:

- The chatbot can provide medicines and insulin individually to diabetes patients in Bangladesh. By message analyzing, the chatbot can know the age, health situation, and other necessary information of the patients. Thus it can provide necessary medicines and health care to diabetes patients.
- This chatbot system will also make an appointment with the nearby doctors for the patient. Patient or user will choose their doctor, and the chatbot can make an appointment with the chosen doctor. This system also can make a direct video call with the doctor if the doctor is available.
- The chatbot system will provide an ambulance in an emergency situation for diabetes patients. It can know or locate the patient's home address and provide a nearby diabetes center or hospital's ambulance so early. Thus the patient can get emergency health services in the nearby hospital.

## 5.10 Conclusion

Intelligent chatbot system is very useful and most effective in every servicing sectors and medical sectors now a days. Now a days diabetes is a very common disease in Bangladesh and a large amount of people are now fighting with this disease in Bangladesh. This study propose an intelligent chatbot system which will help the diabetes patients in Bangladesh by suggesting them nearest diabetes centers, hospitals and doctors. This chatbot system will also help the patients by providing them all kinds of necessary information about diabetes and diet plans. Thus the patients can get the necessary diabetes help properly and can maintain a healthy lifestyle. The survey and results show that the study is quite successful and the chatbot system can provide the necessary services properly to the patients. In the future, the chatbot system will provide necessary medicines to individual diabetes patients and can provide ambulances in emergency cases. It will also have the ability to make an appointment with the doctors for the patients in the future.